

Using the Apache RewriteRule for Clean and Flexible Multilingual URLs

This article shows you how to keep your multilingual websites clean and flexible while still serving them in a way that search engines understand.

Making the contents of your website available in multiple languages is hard enough (unless you are okay with machine-generated junk), but its organisation can sometimes feel even harder. How do I keep the source files organised? How do I keep multiple versions in sync? How do I serve them in a way that search engines understand? And, finally, how do I make the URLs more appealing while still meeting the other criteria and constraints?

There are multiple URL patterns in use for multilingual websites. These are:

- `https://en.example.com/section/subsection/page`
- `https://example.com/section/subsection/page.en.html`
- `https://example.com/section/subsection/page?lang=en`
- `https://example.com/en/section/subsection/page`

First let's see how these work, and what the pros and cons are. Then let's discuss how to choose a scheme that you like even when you are constrained by the limitations of the CMS or site generation method you use.

The differences

The first one depends on the use of subdomains. In this pattern, you are actually running N independent websites, which is highly discouraged due to its complexity and cost. Also, if

a search engine depends on backlinks to rank your site, one version of it might not get the benefits of another even though both serve the same quality content in different languages. So there is no reason to choose this pattern unless you want to keep those versions intentionally independent (e.g., contents differ considerably, regulations, etc).

Well, what about the rest? All of them are fine with respect to SEO and accessibility (as long as you have hreflang, lang and/or Content-Language set).

The `?lang=en` (query param) method is typical if your site is run using a dynamic CMS. The `.en.html` scheme is indicative of a static site generator or manually created files. Both will appear strange to the general audience, compared to the `/en/` prefix scheme.

Choose one, show another

Your CMS or site generator might let you make a selection between these schemes. But what if it doesn't? Also, what if you intentionally want to choose one scheme and expose another? This might be needed if an unappealing scheme is easier to maintain. For example, the `/en/` scheme looks nice,

but it's harder to maintain if your site is directory-based, because you have to replicate and sync the directory structure across all languages (e.g., `/en/section/subsection/page`, `/ml/section/subsection/page`, etc). But if you choose the `.en.html` scheme, you keep all the versions of a page in the same directory, making maintenance much easier. This is where you can rely on your web server to hide what's inside and expose a different interface.

Apache RewriteRule

The Apache RewriteRule helps you transform the URLs that the user requests, either as a visible redirection (so that the URL will change in the browser's address bar), or in a totally transparent manner (so that the URL in the address bar remains the same).

Since writing the correct rules and avoiding redirection loops is tricky, let's try the visible variant first (the user sees the pretty URL becoming the strange one), and then move on to making the process internal.

Here's the RewriteRule snippet that you can put in the `.htaccess` file under your document root (the main directory

